University of Florida, Department of Mathematics

Topology Qualifying Exam, January 2026, Part 1

Be neat, do not write too small. Show your work and support your statements.

1. This problem has two parts.

(a) Let $f, g : X \to Y$ be continuous maps. Define what it means for f to be homotopic to g.

(b) Suppose that the continuous maps $f, g : X \to Y$ are homotopic and that the continuous maps $h, k : Y \to Z$ are homotopic. Show that $h \circ f$ is homotopic to $k \circ g$.

2. If $j : A \hookrightarrow X$ is the inclusion of a retract, then show that the induced map on fundamental groups is injective.

3. This problem has two parts.

(a) State the Seifert–van Kampen theorem.

(b) Determine the fundamental group of $S^1 \vee S^1$.

4. Prove that there are infinitely many distinct homotopy types of spaces with one 0-cell, one 1-cell, and one 2-cell.

5. Let $\mathbb{RP}^2$ denote the real projective plane.

(a) Give the universal cover of $\mathbb{RP}^2$ including the covering map.

(b) What is the group, G, of covering transformations of this cover?

(c) How does it act on the universal cover?

(d) Does $\mathbb{RP}^2$ have a cover, $p : E \to \mathbb{RP}^2$, whose fundamental group, $\pi_1(E)$, is $\mathbb{Z}/(3)$? Explain why or why not.

(e) Does $\mathbb{RP}^2$ have a cover, $p' : E' \to \mathbb{RP}^2$, whose covering transformation group, $G(E')$, is $\mathbb{Z}/(5)$? Explain why or why not.

6. Give the definition of a chain complex of abelian groups and its homology groups.

7. Let (C, d), and (C', d') be chain complexes of abelian groups.

(a) Define what it means for $f : (C, d) \to (C', d')$ to be a morphism of chain complexes, also called a chain map.

(b) For $n \in \mathbb{Z}$ define the induced map $H_n(f) : H_n(C, d) \to H_n(C', d')$.

(c) Prove that $H_n(f)$ is well defined.

8. For $n \geq 0$, let $\Delta^n = \{(x_0, \dots, x_n) \in \mathbb{R}^{n+1} \mid x_i \geq 0,\ \sum_i x_i = 1\}$.

(a) Let X be a topological space. Define the singular chain complex $\mathbf{S}_\bullet(X)$.

(b) Let X be the one-point space $*$. Compute $H_n(X)$ for all n.

9. Use the cellular chain complex to compute the homology of the Klein bottle.

10. This problem has two parts.

(a) State the Mayer–Vietoris sequence for reduced singular homology. You do not need to define the connecting homomorphism.

(b) Let $n \geq 0$. Compute the reduced singular homology groups $\widetilde{H}_k(S^n)$ for all k.